FERA ফেরা

D2B / QUALITY · VERSION 2.0

Technical Requirements · Part B

Security, privacy, costs, evaluation and deployment

Specification handover · Implementation unverified

Authors

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Invented synthetic-data prototype · Interactive team edition

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> Click a section or use the PDF bookmarks. Requirement IDs link to their definitions. External sources are clickable. The meeting workbook contains blank fillable review fields. Application acceptance and pilot evidence remain pending.

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D2B / QUALITY / V2.0

Technical Requirements · Part B

Security, privacy, costs, evaluation and deployment · Authors: Sadman Shaharier and Obidit Islam

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Field	Value
Version	2.0
Date	11 Oct 2026 (Asia/Dhaka, UTC+6)
Owner	Sadman Shaharier (lead/integration)
Authors	Sadman Shaharier; Obidit Islam
Status	V2 team handover; implementation status UNVERIFIED
Split	Part A = §1–§9 (includes §7 schema). Part B = §10–§19. Companion `schema_v2.sql` is a byte-identical copy of Part A §7.
Inputs	D1 and D2 Part A V2.0; validation report; V2 evaluation, privacy, cost and release gates.

V2 release contract

Authors: Sadman Shaharier and Obidit Islam. **Version:** 2.0. **Issued:** 11 October 2026, Asia/Dhaka (UTC+6).

This is the revised team specification, not evidence that the application has been built. Actual repository progress, account configuration, interviews, evaluations, and spend remain **UNVERIFIED** until evidence is attached. V1 defaults are superseded by the V2 rules below; existing requirement IDs are retained.

- This release is a **synthetic-data prototype**. Officer verification and referral delivery are simulated and labelled. The schema rejects records marked non-synthetic; the application must also prevent real reports from entering this release.
- A candidate score is similarity, not identity probability. A reviewer decision does not establish guardian authority or safe reunification. A referral row means **referral recorded**, not delivered or acknowledged.
- Draft payload stays on the device before report confirmation, but providers process audio/text beforehand with a notice. Only expiry/counter/hash metadata and cost reservations are stored server-side before submission.
- Possible duplicates remain visible. Shared phone alone never establishes one child. No physical child-return action is performed by Fera.
- Counts distinguish missing-report rows, found-report rows, and confirmed pairs. They are not a census of unique children or proof of rescue.

- Sources, diagrams, tables, code blocks, and acceptance criteria are part of the handover. The companion code must match the embedded code exactly. Team handover records change closure, evidence gaps, and release gates.

Changes to earlier documents

V2 revises provider notices, pseudonymous redaction, append-only audit privileges, cost reservations, evaluation denominators and a single-database offline topology. Cached demos are excluded from quality results. Existing error IDs remain stable.

10 Security

10.1 Controls

Area	V2 control
Input	Strict runtime schemas on body/query/provider output; NFC; source-span mapping; reject unknown keys and illegal kind/status pairs.
SQL	Parameterised queries; separate migration, runtime and redaction credentials; runtime never owns or alters tables.
Prompt	Text is data, no tools/URL fetching, constrained JSON and supported provenance. Substring presence alone does not validate a fact.
Audio	Tested WebM/MP4/WAV, actual size ≤2 MB and verified duration 5–120 s; codec-aware MIME parsing; Ogg only after verified support/transcoding.
Secrets	Server env, no public keys, no passwords in package; argon2id; secret scanning and committed lockfile.
Paid limits	Server-owned draft counters and HMAC-hashed network keys; atomic budget reservation before dispatch; fail closed for paid calls, keep manual forms available.
Logs	Allowlist request ID, operation/model, duration, costs, success and safe error ID. Never log bodies, transcript, provenance, provider raw errors, names, phones or free notes.
Audit	UPDATE/DELETE/TRUNCATE blocked; safe actor/resource/version/action only. Unknown resource FK null. Rejected transaction audit is written after rollback.
DTOs	Owner/staff-own/officer/admin allowlists; admin rematch returns status only; candidate details officer-only.
CSRF	Same-Origin for all state changes; JSON on standard routes and multipart on transcription; credential cookie and server expiry/revocation checks.
Headers	Test Next.js nonce/hash CSP with hydration/auth/map/poster. Bundled map baseline: self-only images. If OSM tiles enabled, allow canonical tile.openstreetmap.org and required attribution/Referer.
Referrer	Bundled map: no-referrer acceptable. OSM mode: origin-only compatible Referer policy, reviewed privacy notice. Do not suppress required Referer.

Area	V2 control
Demo access	DEMO_ONLY=true; restrict seeded privileged accounts and paid routes. DEMO_MODE fixtures are labelled and never enabled in quality evaluation.
Dependencies	Pinned lockfile, audit and deployment-specific checks; do not claim a dependency scan verifies the application.

10.2 OWASP Top 10 (2021) check

#	Item	Fera exposure	Control	Test
A01	Broken Access Control	Parent or staff reading matched records	Server-side `requireRole`; SQL ownership filters; E-004 for hidden; `verified` read from DB	TC-022, TC-023
A02	Cryptographic Failures	Phones and names in transit or at rest	HTTPS only; managed DB encryption at rest (host-dependent, OQ-021); argon2id passwords; HMAC-signed cookies	Header check in TC-033 run
A03	Injection	SQL; prompt injection via report or import text	Parameterised SQL; data-only prompt framing; schema-only output; evidence check	TC-030, TC-014
A04	Insecure Design	AI auto-confirming matches; alert issuance	No code path from AI to state change; BR-001; no alert feature	TC-010, TC-025
A05	Security Misconfiguration	Debug output, permissive CORS	No CORS (same origin); errors without stack traces; `DEMO_ONLY=true`; protected fixture mode enabled only for labelled demo runs	Manual config review
A06	Vulnerable and Outdated Components	npm supply chain	Lockfile; `npm audit`; pinned major versions	CI audit step
A07	Identification and Authentication Failures	Weak passwords; session theft	Min 10 chars; argon2id; httpOnly Secure cookies; 8 h expiry; brute-force limit	**TC-023**
A08	Software and Data Integrity Failures	Tampered draft token; altered audit	HMAC draft token; append-only audit trigger	**TC-016**
A09	Security Logging and Monitoring Failures	Undetected misuse	Audit of decisions and denied access; pino logs with request IDs	TC-016, TC-022
A10	Server-Side Request Forgery	`source_url` on import	URLs stored only, never fetched by the server	Code review; TC-014

11 Privacy, retention and pilot gate

Only invented synthetic records are permitted in V2. Public availability does not remove a child's privacy interests; real news/alert imports remain outside this release.

System/data	V2 treatment	Real-case gate
Child case fields	Reviewed fields only; no photos, ID documents or precise coordinates	Partner-defined minimisation and lawful basis
Before submit	Payload stays on device; server metadata only	Consent/authority and device threat review
Provider audio/text	Synthetic only, with processing notice before recording/sending	Account/endpoint retention, training, processor and transfer review
Draft storage	Clear on submit/discard; check 24-hour expiry on access; clear-device control	Do not promise physical removal while browser is closed
Audio	In-memory/transit only; not in localStorage or DB	Endpoint terms, actual retention and incident duties
Redaction	Direct identifiers, exact age, source URL, owner/hash/ provenance/vector metadata, outcome/referral/ decision/candidate notes, snapshots and outputs cleared	Field-specific retention and correction/ evidence requirements
Retained record	IDs, age band, sex, district, dates/status and safe audit links under access control	Pseudonymisation; residual linkage/re-identification risk assessment
Found evidence	Redact only after linked missing case is closed	Partner evidence-preservation/correction window
External copies	Downloads, provider copies and backups follow separate policy	Withdrawal/deletion propagation and backup expiry
Map requests	Bundled geometry baseline; optional tiles disclose viewport/request metadata and IP	Approved provider/policy and privacy notice
Poster export	Preview phone/place/age, choose contact inclusion, acknowledge public sharing and no remote recall	Appropriate authority and safer institutional contact
Access	Own scoped views; operational verified-officer evidence; admin metadata	Real officer onboarding, assignment/ supervision and revocation

The checkbox “আমি যাচাই করেছি” verifies accuracy, not legal guardianship. “Nothing shared before confirm” means no submitted-register/candidate sharing; provider processing happens earlier and is disclosed. Retention duration and receiving agency remain partner decisions; they are not invented defaults.

The official Personal Data Protection Act, 2026 is Act 63, dated 10 April 2026, with phased commencement. Its child-data and transfer provisions require qualified assessment before real-case use. Accurate sourced legal references are allowed; the product makes no compliance claim. Official law.

Unpaid Gemini service terms permit improvement/human review uses and caution against personal/sensitive input. OpenAI data treatment varies by endpoint and settings. “No training” is not a universal zero-retention promise. Verify the exact account, model, endpoint and fallback, using Gemini terms and OpenAI data controls. Only synthetic data is allowed until a future release passes the pilot gate.

Before a pilot: responsible agency; approved identity/guardian/safeguarding checks; authorised consent/lawful basis; legal/provider/transfer review; verified onboarding; referral acknowledgement; correction/reopen procedure; retention schedule; access review; breach response; backup/restore test; operational staffing. A data agreement alone does not establish safe operation or national scale.

12 Performance targets and AI budget

12.1 Performance

Target	Value	Measured by
Parent intake first load, mid-range Android, 4G	LCP ≤ 3 s; route JS ≤ 170 KB gzip	Lighthouse mobile profile, 3 runs median
Non-AI API p95	≤ 500 ms (case read, queue, decision, stats)	Playwright timing + server logs
Transcription, 120 s note	≤ 20 s p95 (NFR-003)	`ai_calls.duration_ms`
Extraction	≤ 15 s p95 (NFR-002)	`ai_calls.duration_ms`
Voice note (60 s) → draft shown	≤ 35 s p95 end to end	E2E timer
Matching, one case vs 500 records	≤ 5 s (NFR-004); target ≤ 1 s on 200	`eval/perf.ts`
Map stats, 1,000 cases	≤ 2 s (NFR-005)	`eval/perf.ts`
Concurrent demo users	10 realistic; 50 design (A-020)	k6-style script `eval/load.ts`, error rate < 1 %

12.2 AI usage estimate against USD 30

Provider prices are not in any source: UNKNOWN (OQ-004, OQ-022). The plan below fixes usage volumes and the maximum unit price the budget allows.

Use	Speech (min)	Extraction calls	Basis (Assumption A-037)
Development and testing	60	300	3 devs, 9–13 Oct
WER evaluation	90	0	60 unique clips × 30 s × 3 repeated runs; repeats are not new independent samples
Extraction evaluation	0	220	50 reports × 4 runs × 1.1 retries
Matching evaluation (extracted-field mode)	0	400	200 records × 2 runs

Use	Speech (min)	Extraction calls	Basis (Assumption A-037)
Public import	0	50	Should only
Timing study + rehearsals + video + finale	40	130	5 testers, 10 rehearsals, 3 takes
Total	**190 min**	**1,100 calls**	

Cost line	Allocation	Max unit price that fits
Speech	USD 5	≤ USD 0.026 per audio minute
Extraction (≈ 1,200 input + 400 output tokens per call → 1.32 M in, 0.44 M out)	USD 15	≤ USD 4.87 per 1 M input-equivalent tokens, counting output at 4 × input
Embeddings	USD 0	Local model (DECISION-006)
Reserve	USD 10	Re-runs, API embeddings if RISK-015 fires
Total	**USD 30**	Atomic reservation stop at USD 25; raising toward the USD 30 ceiling requires a documented team budget decision

Rule: at the 11 Oct contract gate, Obidit records real prices from provider pages in `docs/costs.md`. If prices exceed allocation, reduce repeated runs or optional import, retain independent sample coverage and failure reporting, and recalculate actual input/output rates separately. 1.32 + 4×0.44 = 3.08M input-equivalent tokens; 15/3.08 = USD 4.87/M. A reservation upper bound covers maximum tokens/duration; uncertain calls are not refunded silently.

13 Error catalogue

ID	Key	User message (Bangla / English)	Developer log	Recovery
E-001	VALIDATION_FAILED	কিছু তথ্য ঠিক নেই। চিহ্নিত ঘরগুলো দেখুন। / Some details need fixing. Check the marked fields.	`warn` route, Zod issue paths	Highlight fields; keep input
E-002	UNAUTHENTICATED	আবার সাইন ইন করুন। / Please sign in again.	`info` route	Redirect to SCREEN-008, return after
E-003	FORBIDDEN_ACTION	এই কাজের অনুমতি আপনার নেই। / You don't have permission for this action.	`warn` user ID, role, route	Show SCREEN-016
E-004	NOT_FOUND	তথ্যটি পাওয়া যায়নি। / We couldn't find that.	`warn` + `access.denied` audit if signed in	SCREEN-016 with link home

ID	Key	User message (Bangla / English)	Developer log	Recovery
E-005	ILLEGAL_TRANSITION	কেসটির অবস্থা বদলে গেছে। পাতাটি আবার খুলুন। / This case has changed. Reload the page.	`warn` from→to, case ID	Reload latest state
E-006	SUBMISSION_PAYLOAD_CONFLICT	এই জমা নম্বরের তথ্য বদলেছে। তথ্য যাচাই করে আবার জমা দিন। / This submission key has different details. Review before submitting again.	`warn` safe request/key conflict, no payload	Preserve draft; review new submission action
E-007	RATE_LIMITED	{s} সেকেন্ড পরে আবার চেষ্টা করুন। / Please try again in {s} seconds.	`warn` key, count	Countdown, then enable button
E-008	AUDIO_TOO_LARGE	রেকর্ডিং অনেক বড়। ২ মিনিটের কম রেকর্ড করুন। / The recording is too large. Keep it under 2 minutes.	`info` bytes	Re-record or type
E-009	AUDIO_LENGTH_OUT_OF_RANGE	রেকর্ডিং ৫ সেকেন্ড থেকে ২ মিনিটের মধ্যে হতে হবে। / Recordings must be 5 seconds to 2 minutes.	`info` duration	Re-record
E-010	SPEECH_UNINTELLIGIBLE	কথাগুলো পরিষ্কার শোনা যায়নি। আবার বলুন বা লিখে দিন। / We couldn't hear that clearly. Try again or type instead.	`info` transcript length, script ratio	Re-record or focus text box
E-011	SPEECH_UNAVAILABLE	এখন ভয়েস কাজ করছে না। লিখে জানান। / Voice isn't working right now. Please type.	`error` provider, status	Text path
E-012	LLM_INVALID_OUTPUT	সহকারী তথ্য গুছাতে পারেনি। নিচের ফর্মটি পূরণ করুন — আপনার লেখা রাখা আছে। / The assistant couldn't organise this. Please fill the form — your text is kept.	`error` provider, Zod issues (no text)	SCREEN-005 pre-filled
E-013	LLM_UNAVAILABLE	সহকারী এখন সাড়া দিচ্ছে না। ফর্মে তথ্য দিন। / The assistant isn't responding. Please use the form.	`error` both providers, status	**SCREEN-005**
E-014	AI_TIMEOUT	অনেক সময় লাগছে। আবার চেষ্টা করুন বা ফর্ম ব্যবহার করুন। / This is taking too long. Try again or use the form.	`warn` operation, 25 s	Retry once, then SCREEN-005
E-015	UNSUPPORTED_LANGUAGE	বাংলা বা ইংরেজিতে বলুন বা লিখুন। / Please speak or write in Bangla or English.	`info` script ratios	Re-enter
E-016	OUTPUT_BLOCKED	এই লেখাটি তৈরি করা যায়নি। আবার চেষ্টা করুন। / We couldn't create this text. Please try again.	`error` output type, matched term ID	Regenerate; report to team
E-017	REFERRAL_REQUIRED	রেফারেল ছাড়া কেস বন্ধ করা যাবে না। / A referral is needed before closing.	`warn` case ID	Confirm a match first

ID	Key	User message (Bangla / English)	Developer log	Recovery
E-018	EDIT_LOCKED	মিল নিশ্চিত হওয়ার পর কেস বদলানো যায় না। / This case can't be edited after a match is confirmed.	`info` case ID, status	Use the stated case-support process; 13219 remains the external MUN helpline, not a direct Fera-officer contact
E-019	SESSION_EXPIRED	আপনার খসড়ার সময় শেষ। নতুন করে শুরু করুন। / Your draft has expired. Please start again.	`info` session age	Start new intake
E-020	OUTPUT_REQUEST_LIMIT_RESERVED	Reserved for future output-abuse limits; not an active V2 regeneration counter. Identical current-version templates are reused.	Safe rate-limit metadata only	Reuse current output; no five-generation rule
E-021	INTERNAL	কিছু একটা ভুল হয়েছে। আবার চেষ্টা করুন। / Something went wrong. Please try again.	`error` stack (server only), request ID	Retry; draft kept
E-022	MIC_UNAVAILABLE	মাইক্রোফোন চালু করা যায়নি। আপনি লিখে জানাতে পারেন। / The microphone isn't available. You can type instead.	client `console.warn` only	Focus text box
E-023	AI_BUDGET_CAP	সহকারীর সীমা শেষ। বাকি তথ্য ফর্মে দিন। / The assistant's limit is reached. Add the rest in the form.	`warn` calls used or spend total	SCREEN-005
E-024	NETWORK_OFFLINE	ইন্টারনেট সংযোগ নেই। আপনার তথ্য রাখা আছে — সংযোগ ফিরলে আবার পাঠান। / No internet connection. Your details are saved — send again when you're back online.	client only	Retry button; draft kept in browser

14 Observability

Item	Specification
Logs	pino JSON to stdout: `time`, `level`, `request_id`, `route`, `role`, `status`, `duration_ms`, `error_id`. PII redacted (§10.1).
AI metrics	One `ai_calls` row per call: operation, provider, model, duration, units, estimated cost, cache hit, error ID.
Spend report	`npm run spend` prints today's and total spend vs `AI_SPEND_CAP_USD`. Run daily (TASK-059).
Health	API-019 `/api/health`: DB ping, embedding model loaded, demo mode flag, git SHA.
Debug mode	Verified officers may inspect safe matching components on SCREEN-010; admins receive request/model/cache operational metadata only, never candidates, scores or child payload. Debug does not bypass DTO/role checks.

Item	Specification
Demo-safe fallback	`DEMO_MODE=1` loads `/fixtures/demo/*.json` (transcripts, extraction outputs and embeddings for 3 scripted cases). Lookup key: SHA-256 of normalised text, or `fixture_id` for the 3 scripted audio files. Any cached result shows a small "ক্যাশ" (cached) badge so judges are never misled.
Alerts	None automated. Lead checks `/api/health` and `npm run spend` before each demo.

15 Deployment

15.1 Environment variables

Name	Example	Required
`DATABASE_URL`	`postgres://fera:…@host:5432/fera?sslmode=require`	✓
`AUTH_SECRET`	32+ random bytes, base64	✓
`PARENT_COOKIE_KEY`, `DRAFT_SIGNING_KEY`	32+ random bytes each	✓
`LLM_PROVIDER`, `LLM_MODEL`	`gemini` + model ID from OQ-004	✓
`LLM_FALLBACK_PROVIDER`, `LLM_FALLBACK_MODEL`	second provider	✓
`GEMINI_API_KEY` / `OPENAI_API_KEY` / `ANTHROPIC_API_KEY`	as held	per provider
`STT_PROVIDER`, `STT_MODEL`	`whisper` + model ID	✓
`EMBED_PROVIDER`, `EMBED_MODEL`, `EMBED_CACHE_DIR`	`local`, candidate model, `./models`	✓
`AI_SPEND_CAP_USD`	`25`	✓
`DEMO_ONLY`	`true`; blocks real-case intake	✓
`DEMO_MODE`	`0` for quality evaluation; `1` only for labelled scripted fixtures	✓
`SEED_ADMIN_USER`, `SEED_ADMIN_PASSWORD`	for seeding only	seed

15.2 Local setup (≤ 15 min working target, NFR-014)

The following commands are an implementation target, not delivered application code. The handover contains specification companions only. Apply the schema to a fresh isolated DB; define the npm scripts and README in the actual repository, then test setup time.

```
git clone <repo> fera && cd fera
cp .env.example .env.local            # fill keys
```

```
docker compose up -d db               # postgres 16 + pgvector image (pin and verify exact image/
extension versions at contract gate)
psql "$DATABASE_URL" -v ON_ERROR_STOP=1 -f code/schema_v2.sql
npm ci
npm run model:fetch                   # downloads embedding model to ./models
npm run seed -- --set demo            # districts, neighbours, users, ~200 synthetic records
npm run dev                           # http://localhost:3000
```

15.3 Build and deploy

Step	Command / action
Check	`npm run lint && npm run typecheck && npm test`
Build	`npm run build`
Deploy (Round 2)	Push to `main` → Vercel production deploy. Preview deploy per pull request.
Database	Apply `schema_v2.sql` once on the managed host; seed with `npm run seed -- --set demo` using the production URL.
Smoke test	`npm run smoke -- --base https://<prod-url>` (health, sign-in, one extract, map stats)
Finale laptop	`DEMO_MODE=1 npm run build && npm start`; local DB seeded; phone uses the deployed HTTPS URL over mobile data (A-031).

15.4 Rollback

Layer	Method
App	Re-deploy the previous green commit; tag `r2-freeze` on 14 Oct and `finale` on 23 Oct.
Schema	`schema_v2.sql` is version 1.0 only. Any change ships as `migrations/NNN_*.sql` with a matching down script.
Data	`npm run seed -- --reset --set demo` restores demo data in < 2 min.

15.5 Seed and demo data

Set	Contents
`districts`	64 rows with Bangla/English names and centroids (DEP-004)
`neighbours`	Team-compiled adjacency (A-013)
`users`	1 admin, 2 verified officers, 1 unverified officer, 2 staff — seeded passwords from env
`synthetic`	~200 records, all `is_synthetic=true`, incl. 3 named demo cases (D6 §9)
`fixtures/demo`	Cached outputs for the 3 scripted inputs

16 Trade-offs

Decision	Alternatives	Reason	Cost	Migration path
DECISION-004 LLM adapter, extraction only	Single vendor SDK throughout	No lock-in; fallback on outage; pick best Bangla extractor after eval	~4 h extra code; 3 prompt variants to test	Add a provider class; set env
DECISION-002 pgvector in Postgres	Separate vector DB service	One store, one backup, transactional with case state; 200–30,000 rows is small	HNSW tuning limited; vector search shares DB CPU	Export embeddings; swap COMP-012 query layer
DECISION-008 Embeddings + rules	Pure LLM pairwise judging	Deterministic, explainable, offline, ~0 cost; LLM over 200×200 pairs breaks budget and NFR-004	Misses some semantic nuance	Add an optional LLM re-rank of top 5 behind a flag
DECISION-006 Local embeddings	API embeddings	Free, offline for finale	Model size/cold start risk (RISK-015)	`EMBED_PROVIDER=api` + `npm run reembed`
DECISION-007 Template outputs	LLM-written GD, poster, reasons	Cannot add facts (BR-022); offline; 0 calls	Less natural wording	Add LLM "polish" behind flag with fact-diff check
DECISION-010 Deployed app + local copy	Deployed only	Venue Wi-Fi risk	Two environments to seed	Single env after the hackathon
DECISION-016 Dev/ test split for matching evaluation	Tune and score on the same set	Avoids inflated metrics (RISK-007)	Smaller test set	Larger set later
DECISION-017 Hard spend cap in app	Watch provider dashboards only	Budget is fixed at USD 30	One table + one check	Raise cap via env

17 Evaluation design and reporting

Metric	Set	Definition	Script
Extraction GOAL-002	50 unique labelled reports; identity/variant-grouped dev/test; exact source spans	Present-field accuracy, false additions/ omissions, per-field precision/recall; null agreement reported separately, never headline accuracy. Phone exact, exact time tolerance separate from approximate interval correctness.	eval:e xtract
Matching GOAL-003	Manifest: 80 missing + 80 found + 40 additional decoys = 200 total; 60 canonical true pairs, 20 canonical no-match missing queries; 25 duplicate relationships within the same 200	Identity/duplicate-connected-group split 40% dev/60% test; all variants stay together. Hit/ Recall@5 on true-pair queries; top-1 accuracy and coverage; query-level P@5 and separately named aggregate shortlist precision.	eval: match

Metric	Set	Definition	Script
No-match and uncertainty	Include unknown age, overlapping times, shared contacts, wrong district, similar names and contradictions	Report false-candidate/above-threshold suggestion rates, abstention coverage, true-pair misses; minimum qualification threshold frozen on dev.	eval: match
Duplicates GOAL-004	25 declared same-identity duplicate links; explicit different-child shared-phone negatives	Pairwise precision/recall/F1 plus false consolidation/hiding count. Detection only; no claimed merge accuracy.	eval:d edupe
Speech GOAL-005	20 prompts × 3+ speakers = 60 unique clips; noise/dialect/Banglish slices; repeated runs labelled	WER=(S+D+I)/N may exceed 100%; tokenisation documented; unique count distinct from repeats; phone/age/place/time errors and speech→extraction errors.	eval:w er
Timing GOAL-001	5+ adult volunteers, invented scenarios, paired/counterbalanced methods	Median/range and completion numerator/ denominator; help and order effects recorded; five-person 90% threshold requires 5/5. Exploratory, not population efficacy.	eval:ti me
Human/ audit GOAL-008	Known seeded lifecycle plus concurrent and denied actions	Every terminal coordination event has a verified recorded decision where needed and a safe audit; identity verification is not inferred.	eval:a udit

Manifest fields: record_id, kind, identity_id, duplicate_group_id, true_partner_ids, no_match, decoy_type, source_variant, split, speaker_id where applicable, and ground_truth_revision. Assert total=200 and declared link/query counts. The 40 decoys are additional to the 160 canonical reports; duplicate variants/ links are included within the 200, not an extra uncounted set. Group splitting may alter the exact 40/60 count: report actual counts rather than force identity leakage. Validate an independent test manifest before any tuning.

Ground truth is written and hashed before matcher tuning. Tune prompts/weights/thresholds only on dev. Freeze model revision, preprocessing, embedding space, score components, qualification/dedupe thresholds, prompt/template hashes, and goals before held-out runs. Evaluation enforces DEMO_MODE=0 and cache_hit=false; a fixture hit invalidates the relevant quality run.

Baselines: manual form timing; rules-only extraction where feasible; rules-only matching; embeddings-only and combined ranking. Run gold-field and speech/extracted-field modes. Separate live AI latency/ spend from cached demo timing. Record model/account mode, sample/query count, SHA, date, hardware/deploy, denominators, uncertainty and at least two failures. Approximately 36 true-pair queries may remain from 60 pairs at a 60% split; do not claim national accuracy from this small synthetic test.

Synthetic record text, outcome labels and fixture names do not imply successful rescue or operational officer adoption. If a promised metric cannot be measured, show “not measured” with reason and next action; never a target as an achieved value.

18 Technical acceptance criteria

#	Criterion	Check
1	Every Must FR passes its TC on the protected synthetic deployment; unmet items are disclosed and block the claimed feature.	Playwright run report
2	`schema_v2.sql` applies cleanly to an empty Postgres 16 + pgvector database.	CI job
3	Illegal state transitions fail at the DB trigger even when called directly.	**TC-032**
4	`audit_log` UPDATE, DELETE and TRUNCATE fail; runtime role cannot alter/disable protections.	**TC-016**
5	Parent/staff responses contain no candidates or other-party found-case fields; staff own-entry and parent own-edit DTOs are tested.	TC-019, TC-022
6	Logs from a full E2E run contain 0 phone numbers and 0 child names (regex scan).	**TC-028**
7	Scripted journey completes with network disabled in `DEMO_MODE=1`.	**TC-021**
8	`npm audit --omit=dev`: 0 high/critical.	CI
9	axe plus manual keyboard, Bangla screen reader, short-cut disabling, 320px reflow/ 400% viewport and 200% text zoom; no unresolved AA issue in required flows.	**TC-033**
10	Atomic cap defaults USD 25; any reviewed increase stays ≤ USD 30; reservations and actual settled spend are reported.	`npm run spend`
11	All §12.1 targets met or the miss reported in the deck.	`eval/perf.ts`
12	Evaluation report contains all 5 deck metrics with failure examples.	**TC-020**

19 Engineering task breakdown

TASK IDs, owners, dependencies and done criteria are in D6 §5. The full board totals 222 h; the specified Round 2 subset totals 184 h. Actual remaining effort requires a completed-work inventory.

Area	TASKs	Hours
Setup and spikes	TASK-001 – TASK-005	11
Frontend	TASK-008, TASK-021, TASK-022, TASK-024, TASK-025, TASK-026, TASK-027, TASK-028, TASK-029, TASK-030, TASK-031, TASK-032	47
Backend	TASK-006, TASK-007, TASK-015, TASK-020, TASK-023, TASK-059	24
Database and data	TASK-009, TASK-010, TASK-011, TASK-019	16
AI pipeline	TASK-012, TASK-013, TASK-014, TASK-016, TASK-017, TASK-018, TASK-033	31

Area	TASKs	Hours
Integration and polish	TASK-034, TASK-035, TASK-036	8
Testing and evaluation	TASK-037 – TASK-041, TASK-043, TASK-060	24
Deployment	TASK-042, TASK-053	6
Demo and submission	TASK-044, TASK-045, TASK-046, TASK-054, TASK-055, TASK-058	20
Should / Could	TASK-047 – TASK-052, TASK-056, TASK-057	23
V2 contract, schema, budget and manifest remediation	TASK-061 – TASK-064	12
All tasks	**64 tasks; full task-board estimate**	**222**

Assumptions (Part B)

ID	Assumption
A-037	Usage volumes in §12.2 (190 audio minutes, 1,100 extraction calls, ~1,200 input + 400 output tokens per call).
A-038	Output tokens cost about 4 × input tokens for budgeting; replace with verified real prices at the 11 Oct contract gate.
A-039	The canonical manifest totals 200 reports: 80 missing + 80 found + 40 decoys; 60 canonical true pairs, 20 no-match missing queries and 25 duplicate links within that same total.

IDs introduced

- E-024 NETWORK_OFFLINE (client only)
- DECISION-016 Dev/test split for matching evaluation
- DECISION-017 Hard AI spend cap in app
- A-037 AI usage volumes
- A-038 Output/input token price ratio
- A-039 Synthetic set composition

IDs referenced

- E-001 – E-023, DECISION-002, DECISION-004, DECISION-006 – DECISION-008, DECISION-010, API-019, COMP-012, COMP-021 (D2 Part A)
- FR-025, BR-001, BR-006, BR-011, BR-022, NFR-002 – NFR-005, NFR-014, NFR-016, A-010 – A-013, A-020, A-031 (D1, D2 Part A)

- GOAL-001 – GOAL-005, GOAL-008 (00)
- OQ-004, OQ-011 – OQ-013, OQ-021, OQ-022 (00, D1, D2 Part A)
- RISK-007, RISK-015 (D1, D2 Part A)
- DEP-004 (D1)
- SCREEN-005, SCREEN-008, SCREEN-010, SCREEN-016 (D1)
- TC-010, TC-014, TC-016, TC-019 – TC-023, TC-025, TC-028, TC-030, TC-032, TC-033 (D1)
- TASK-001 – TASK-060 (D6)

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